

EE5239 Nonlinear Optimization HW 5 Cover Sheet

Instructor name: Mingyi Hong, Jiaxiang Li

Student name: _____

- Date assigned: Wed. 11/13/2024
- Date due: **Wed. 11/27/2024** Midnight
- This cover sheet must be signed and submitted along with the homework answers on additional sheets.
- By submitting this homework with my name affixed above,
 - I acknowledge that I am aware of the University’s policy concerning academic misconduct (appended below),
 - I attest that the work I am submitting for this homework assignment is solely my own, and
 - I understand that suspiciously similar homework submitted by multiple individuals will be reported for investigation.
- Academic Misconduct in any form is in violation of the University’s Student Disciplinary Regulations and will not be tolerated. This includes, but is not limited to: copying or sharing answers on tests or assignments, plagiarism, having someone else do your academic work or working with someone on homework when not permitted to do so by the instructor. Depending on the act, a student could receive an F grade on the test/assignment, F grade for the course, and could be suspended or expelled from the University.

1 Reading

- Reading: Textbook Section 3.3.1. 3.3.4, Section 3.4 (**Second edition book**)
- Reading: Textbook Section 4.3.1. 4.3.2 (subsection, sufficient conditions and Lagrangian Minimization), Section 4.4 (**Third edition book**)

2 Note

Please note that besides regular written problems and programming problems, you need to choose one more problem from either **Option 1**, which is one more theoretical problem, or **Option 2**, which is a programming problem.

3 Written Problems

1. (40 points) Exercise 3.3.1, Exercise 3.3.3, Exercise 3.4.1, Exercise 3.4.3 (version 2 of the book),

or Exercise 4.3.1, Exercise 4.3.3, Exercise 4.4.1, Exercise 4.4.3 (version 3 of the book)
[**Note, for 3.3.3/4.3.3, only need to solve. the case where $N = 2$**]
2. (**Option 1**) (20 points) Exercise 3.4.4 (version 2 of the book),

or Exercise 4.4.4 (version 3 of the book)

4 Coding Problems

1. (40 points) This assignment is a follow-up on HW 3, 4. We will apply the algorithms we learned so far to classify the hand writing digits. The homework is quite open ended, the objective is to give you some insights on the practical performance of different algorithms and formulations that we have learned so far. The data sets are available at the course web site.

The problem: Utilizing the two features that are provided by the data set, perform a few classification tasks on the data set using optimization algorithms. You can define your own tasks, for example separating digit “1” and “5”, or separating “1” and the rest.

Use **support vector machine** (SVM) to perform the classification. Pick **two algorithms** that we have learned so far (coordinate descent, gradient descent, different stepsizes, etc). Once a model is built by a chosen model/algorithm, please plot the separating line graphically as is done in the slides. Please also test the performance of different algorithms using **test data set** (i.e., the percentages of error), and compare with the models generated by the previous HWs; see Lecture 5 slides.

The requirement: A report needs to be written on the work that has been performed. Important parts that need to be covered are: 1) description of the model; 2) description of the algorithm that solves the model; 3) description of the results.

Other requirements are similar to HW 4.

Please put your codes in a zipped folder and upload to Canvas.

Also upload a separate pdf file to the Canvas.

The data sets: Please download from the course web site the “handwriting data.zip” folder. Please first read the “info.text” to get an idea of the overall data set. The data files that you are going to use are “feature_test.txt” (testing data) and “feature_train.txt” (training data). As have been explained before, here two specific features have already been computed for you (“symmetry” and “intensity”), for each of the data point.

2. **(Option 2)** (20 points) Perform the same task as above, but use the **raw data with full feature (256 features now)**. Compare your results with your existing classification results and discuss.